



SCI_BUILD_A2_W06 · BUILD Science teacher note

Staff-facing; review only. Verbatim SoW objective: Plan and carry out a simple fair test on rocks.

Source: BUILD scheme of work, Autumn 2 week 6. Pupil goal: I am learning to keep a rock test the same each time.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can name what we change., I can name what stays the same., I can record what I see.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	Name one raw and one made item.	Raw rock and finished brick.	BUILD scheme of work, Autumn 2 week 5
2-4 weeks	Why can slate suit a roof?	A suitable slate can form thin roof pieces.	BUILD scheme of work, Autumn 2 week 4
last term	How can food help a body?	It gives nutrition.	BUILD scheme of work, Autumn 1 week 6

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: Do three drops on A and ten on B keep the water amount the same? Answer: No. Use the same amount for both. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
fair test	Tier 3	a check with one planned change	I use three drops each time.
same	Tier 2	not changed in each try	I use the same water.
change	Tier 2	make one thing different	I change the rock.
result	Tier 3	what the check showed	A drop stayed on top.

Faded example, tiered task and stretch

I do full: "A gets three drops. I wait one count. B gets the same." We do part-done: "A gets three drops. B gets drops. I wait the same time." Removed step: Fill three as the water amount for rock B. Independent: no filled step. Next fade: Remove the amount and the time prompt; pupil plans both controls.

We do F: Place three dot counters by each rock. M: Plan what stays the same and what changes. H: Why could different sample sizes spoil this check? Independent F: Point to what stays the same. M: Plan one fair water check. H: One rock is much larger. How could you compare? Go deeper: Drops run off an edge. Did water pass through? Fix it. Named harder stretch — Fairness repair: Find two unfair steps. Fix them.

The kit route is hands-on or sensory: two safe rocks, dropper, tray and towel, dot counters. Two on-screen interactions are card sort and What should we put in each sample?; native keyboard



controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Practical: Predict first: Will the drops act the same? Kit: Two like-size safe rocks, Dropper, Tray.
Steps: Guess first; Add three drops to A. Wait one count; Do the same on B; Record and compare.
Results headings: Rock, Drops, What happened? Check against the school risk assessment and CLEAPSS guidance. Use clean blunt samples; control the water and dry any spill.

Hinge and decision

Question: New samples C and D get four and nine drops. Is the water amount controlled?

Option	Correct?	Diagnosis
No. Give each four drops.	yes	Keeps the water amount equal across samples.
Yes. Both got some water.	no	Ignores the different amounts.
No. Use the same rock.	no	Removes the changed sample from the comparison.
Yes. D could be bigger.	no	Adds an unsupported reason for unequal amounts.

Rule: About 3 in 4 or more right → move on; otherwise re-model A gets three drops. I wait one count. B gets the same. Ask the parallel hinge. No scores or answers saved. Parallel hinge: New rocks E and F each get four drops, but E waits ten counts and F waits two. Is the wait fair? No; keep the wait the same.

Misconceptions:

Misconception	Spot	Correct
Any water on both samples makes the test fair.	Accepts three drops versus ten.	Use the same amount for both while the sample changes.
The two tests must use the same rock to be fair.	Refuses to change sample.	Change the rock you compare; keep water amount and time the same.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ? At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ? R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations

Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Give a three-step visual card for keep a rock test fair; remove its last cue after a correct try.
Communication & Interaction	explicit instruction	Point to the real kit before a short spoken, signed or AAC response; wait for a turn.
SEMH	cognitive/metacognitive strategies	Offer a private first-then cue and planned pause before the hinge; re-enter with one step.



Broad area	EEF five-a-day strategy	Specific support
Sensory/Physical	technology	Use a stable tray, large cards or an accessible input; no forced touch.
EAL (language, not SEN)	flexible grouping	Rehearse a full M or H idea in any language before writing; keep the science challenge.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the object or model. → Ask: What will you check next? → Model one step and give the next turn back. → Finish one example together, then let the pupil try. No physical prompt. Stage scripts: I do “Watch my first step. What do you see?”; We do “What step is missing? Show it with the kit.”; Independent “What can you do first on your own?”. At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: Point to the dropper. → The tool used to add drops; What stays the same? → The drop count and wait time.
- M: Are three drops and ten drops fair? → No. Give each rock the same drop count; What is a result? → What the check showed.
- H: One sample is much larger. What will you do? → Use samples of similar size or state that size also changed; How could run-off fool you? → A drop in the tray may have run off the side, not through the rock.

Regulation: Agree a quiet pause and supervised place. Give one short cue. Re-enter at the last completed step. No evidence while distressed.

Lesson record: BUILD_SCI_A2_W06 via staff device or projector only. Gatsby 4: A lab worker keeps one test step the same so results can be compared.

Reading gate: BUILD below age 8; rough ARI grade 0.38 on 647 pupil-visible words (adult guidance and navigation excluded); heuristic age guide 6.38 years, not a formal reading-age test. This is a rough screen, not an individual assessment.

Pupil-facing risk: Rock edges may scratch skin; water may spill.

Pupil-facing safety: Use blunt rocks in a tray. Wipe spills. Do not taste; wash hands after.

SLT planning and evaluation

Lesson code: SCI-BUILD-A2-W06

SoW reference: BUILD scheme of work, Autumn 2 week 6

Exact SoW outcome: Plan and carry out a simple fair test on rocks.

Evidence (scheme of work): Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.



Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-BUILD-A2-W06. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): See two rocks and a dropper. Staff: Offer a safe kit choice and short first-then cue.

Starter (3 min): Think back. Say, sign or point. Staff: Ask three low-stakes prior questions; answers in note.

I Do (5 min): Watch a fair water check. Staff: Pre-teach words; show the full model; ask the show-me check.

We Do (7 min): Plan and run the same steps. Staff: Hand over the missing step; offer F or H route. Reset once.

We Do 2 (5 min): All of us show one choice. Staff: Take all pupil responses at once; use decision rule and parallel hinge.

Independent (8 min): Test two rocks. Record what you see. Staff: Choose one adult for supported pupils, one for group; swap across the week.

Review (4 min): Check two rows. Hear a next step. Staff: Give VF and one NS(+); invite one self-edit. Reset once.

Exit (3 min): Say what stayed the same. Staff: Hear feedback and next step, choose next start, then one-minute reset.

Planned adaptation — cognition learning: Give a three-step visual card for keep a rock test fair; remove its last cue after a correct try.

Planned adaptation — communication interaction: Point to the real kit before a short spoken, signed or AAC response; wait for a turn.

Planned adaptation — semh: Offer a private first-then cue and planned pause before the hinge; re-enter with one step.

Planned adaptation — sensory physical: Use a stable tray, large cards or an accessible input; no forced touch.

Planned adaptation — eal: Rehearse a full M or H idea in any language before writing; keep the science challenge.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: explicit instruction; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: flexible grouping

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Test two rocks. Record what you see. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W07. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W07: Review and evidence learning on skeletons, nutrition & rocks (Entry Level Science / UAS).

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:



Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Fill three as the water amount for rock B. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "New samples C and D get four and nine drops. Is the water amount controlled?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: See two rocks and a dropper. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Test two rocks. Record what you see. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use fair test, same, change, result with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.